

Discrete Mathematics

Week 2 – Sets, Functions & Matrices

JeYoung Park

SeoulTech CS&CE

2026-03-26

Today's Roadmap

Part 1: Sets

- 1 Set – Definition & Notation
- 2 Subset, Power Set
- 3 Cartesian Product
- 4 Set Operations & Identities
- 5 Bit Representation & C++
- 6 Multiset, Countability

Part 2: Functions & Matrices

- 7 Functions
- 8 Matrix – Definition & Operations
- 9 Boolean Matrix
- 10 Practice Problems

CS Motivation

Set = type system, DB, bitmask

Matrix = graph, relation, network

Week 1 Review & Week 2 Bridge

Last Week

Proposition 과 논리 연산자로 참/거짓을 판별하는 언어를 배웠습니다.

$p \wedge q$, $p \vee q$, $\neg p$, $p \rightarrow q$

This Week: Proposition $\rightarrow$ Set

Week 1 (Logic)		Week 2 (Set)
$p \wedge q$	$\longleftrightarrow$	$A \cap B$
$p \vee q$	$\longleftrightarrow$	$A \cup B$
$\neg p$	$\longleftrightarrow$	$\bar{A}$ (complement)
$p \rightarrow q$	$\longleftrightarrow$	$A \subseteq B$

Why Sets Matter in CS

C++의 `int`, `bool`, `char` 같은 데이터 타입은 집합 위에 정의됩니다.

예: `boolean`은 집합 $\{0, 1\}$ 에 AND, OR, NOT 연산을 붙인 것입니다.

Set = 수학적 데이터 구조의 기반, **Matrix** = 관계를 숫자로 표현하는 언어

Set – Definition & Notation

Definition

Set은 순서 상관 없이 모인 서로 구별되는 객체(elements / members)의 모음입니다.

Notation

- **Roster method:** $A = \{1, 2, 3, 4\}$
- **Set-builder notation:**
 $B = \{x \mid x \text{ has property } P\}$
“the set of all x s.t. x has property P ”
e.g. $\{x \mid x \text{ is a positive even}\} = \{2, 4, 6, \dots\}$
- $2 \in A$ (element of)
- $5 \notin A$ (not element of)

Well-known Sets

- $\emptyset$ empty set
- $\mathbb{N}$ $\{0, 1, 2, \dots\}$
- $\mathbb{Z}$ $\{\dots, -1, 0, 1, \dots\}$
- $\mathbb{Q}$ rationals
- $\mathbb{R}$ reals
- U universal set

REMARK

Order does not matter: $\{1, 2, 3\} = \{3, 1, 2\}$

No duplicates: $\{1, 1, 2\} = \{1, 2\}$

Equality: $A = B \Leftrightarrow \forall x (x \in A \leftrightarrow x \in B)$

Datatype = Set + Operations

Definition

Datatype (Type) = 집합의 이름 + 그 집합 위에 정의된 연산(operation)들

C++ Type	Underlying Set	Operations
bool	$\{0, 1\}$ (false, true)	&&, , !
char	$\{0, 1, \dots, 255\}$	comparison, arithmetic
int	$\{-2^{31}, \dots, 2^{31}-1\}$	+, -, *, /, %
unsigned int	$\{0, 1, \dots, 2^{32}-1\}$	+, -, *, bitwise ops

Why This Matters

Type system이 집합론에 기반하므로,
후에 **type safety**, **overflow**, **casting**의 동작 원리 또한 Set+operations 에 의존 합니다.

Subset vs Proper Subset

Subset $A \subseteq B$

A의 **모든** 원소가 B에도 포함

$$A \subseteq B \Leftrightarrow \forall x (x \in A \rightarrow x \in B)$$

Proper Subset $A \subset B$

A는 B의 부분집합이지만 $A \neq B$

$$A \subset B \Leftrightarrow (A \subseteq B) \wedge (A \neq B)$$

즉, $\forall x (x \in A \rightarrow x \in B) \wedge \exists x (x \in B \wedge x \notin A)$

B에는 있지만 A에는 없는 원소가 **적어도 하나** 존재

Example

$$A = \{1, 2\}, B = \{1, 2, 3\}$$

$$A \subseteq B \quad \checkmark \quad (1, 2 \text{ 모두 } B \text{에 있음})$$

$$A \subset B \quad \checkmark \quad (3 \text{이 } B \text{에만 있음})$$

$$B \subseteq A \quad \times \quad (3 \in B \text{이지만 } 3 \notin A)$$

$$C = \{1, 2, 3\}$$

$$B \subseteq C \quad \checkmark$$

$$B \subset C \quad \times \quad (B = C)$$

REMARK

모든 집합 S에 대해:

$$\emptyset \subseteq S \quad (\text{공집합은 모든 집합의 부분집합})$$

$$S \subseteq S \quad (\text{자기 자신도 부분집합})$$

Cardinality & Power Set

Cardinality $|A|$

집합 A의 원소 수

$$A = \{a, b, c\} \Rightarrow |A| = 3$$

$$|\emptyset| = 0$$

Finite set: $|A|$ 가 자연수

Infinite set: A set is said to be infinite if it is not finite

e.g. $|\mathbb{N}| = \infty$, $|\mathbb{Z}| = \infty$

In C++

```
std::set<int> s;  
s.size() = |S| (cardinality)  
s.empty()  $\leftrightarrow S = \emptyset$ 
```

Power Set $\mathcal{P}(A)$

$\mathcal{P}(A)$: A의 모든 부분집합의 집합

$$|A| = n \Rightarrow |\mathcal{P}(A)| = 2^n$$

Example: $A = \{a, b, c\}$

$$\mathcal{P}(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \\ \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$$

$$|\mathcal{P}(A)| = 2^3 = 8$$

Why 2^n ?

각 원소에 대해 “포함 / 미포함” 2가지 선택

$$n \text{개 원소} \rightarrow \underbrace{2 \times 2 \times \cdots \times 2}_n = 2^n$$

= **bitmask**로 열거 가능 (n-bit integer)

Cartesian Product

Definition

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$$

A와 B의 원소로 만들 수 있는 모든 **ordered pair**의 집합

Example

$$A = \{1, 2\}, B = \{a, b, c\}$$

$$A \times B = \{(1, a), (1, b), (1, c), \\ (2, a), (2, b), (2, c)\}$$

$$|A \times B| = |A| \cdot |B| = 6$$

REMARK

$A \times B \neq B \times A$ (순서쌍은 순서가 중요!)

$$(A \times B) \times C \neq A \times B \times C$$

$((1, 2), 3)$ vs $(1, 2, 3)$ – 구조가 다름

Exception: $A = \emptyset$ or $B = \emptyset$ or $A = B$

CS Applications

Relation: $R \subseteq A \times B$

e.g. Student–Course enrollment
 $(\text{Alice}, \text{DiscreteMath}) \in R$

DB table: each row = tuple

Graph edge: $(u, v) \in E \subseteq V \times V$

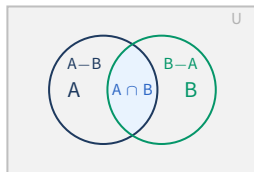
In C++

```
pair<int, char> p = {1, 'a'};  
// p.first=1, p.second='a'  
//  $(1, 'a') \in \text{int} \times \text{char}$ 
```


Set Operations

Operation	Symbol	Definition
Union	$A \cup B$	$\{x \mid x \in A \vee x \in B\}$
Intersection	$A \cap B$	$\{x \mid x \in A \wedge x \in B\}$
Difference	$A - B$	$\{x \mid x \in A \wedge x \notin B\}$
Complement	$\bar{A}$	$\{x \in U \mid x \notin A\}$
Sym. Diff.	$A \oplus B$	$(A - B) \cup (B - A)$

Venn Diagram



Logic $\leftrightarrow$ Set

$$\begin{array}{ll} A \cup B \leftrightarrow p \vee q & A \cap B \leftrightarrow p \wedge q \\ \bar{A} \leftrightarrow \neg p & A \oplus B \leftrightarrow p \oplus q \end{array}$$

Example

$$A = \{1, 2, 3, 4\}, B = \{3, 4, 5, 6\}, U = \{1, \dots, 6\}$$

$$A \cup B = \{1, 2, 3, 4, 5, 6\} \quad A \cap B = \{3, 4\}$$

$$A - B = \{1, 2\} \quad \bar{A} = \{5, 6\}$$

$$A \oplus B = \{1, 2, 5, 6\}$$

Set Identities

REMARK

Week 1의 논리 동치 법칙과 완전히 같은 구조! $\wedge \leftrightarrow \cap$, $\vee \leftrightarrow \cup$, $\neg \leftrightarrow \bar{}$ 로 치환하면 됩니다.

Set Identity	Logical Equiv.	Name
$A \cap U = A$, $A \cup \emptyset = A$	$p \wedge \mathbf{T} \equiv p$, $p \vee \mathbf{F} \equiv p$	Identity
$A \cup U = U$, $A \cap \emptyset = \emptyset$	$p \vee \mathbf{T} \equiv \mathbf{T}$, $p \wedge \mathbf{F} \equiv \mathbf{F}$	Domination
$A \cup A = A$, $A \cap A = A$	$p \vee p \equiv p$, $p \wedge p \equiv p$	Idempotent
$\overline{(\bar{A})} = A$	$\neg\neg p \equiv p$	Double Complement
$A \cup B = B \cup A$, $A \cap B = B \cap A$	Commutative	Commutative
$(A \cup B) \cup C = A \cup (B \cup C)$	Associative	Associative
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	Distributive	Distributive
$\overline{A \cap B} = \bar{A} \cup \bar{B}$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$	De Morgan
$\overline{A \cup B} = \bar{A} \cap \bar{B}$	$\neg(p \vee q) \equiv \neg p \wedge \neg q$	De Morgan
$A \cup (A \cap B) = A$, $A \cap (A \cup B) = A$	Absorption	Absorption
$A \cup \bar{A} = U$, $A \cap \bar{A} = \emptyset$	$p \vee \neg p \equiv \mathbf{T}$	Complement

Why Proofs?

이 항등식들은 “당연해 보여도” 증명이 필요합니다. 방법: (1) **Element-wise proof** 또는 (2) **Logical equivalence chain**

Proof: De Morgan's Law – Logical Equivalence Method

Claim: $\overline{A \cap B} = \bar{A} \cup \bar{B}$

Set-builder notation과 논리 동치를 이용한 증명:

$$\begin{aligned}\overline{A \cap B} &= \{x \mid x \notin A \cap B\} && \text{(definition of complement)} \\ &= \{x \mid \neg(x \in A \cap B)\} && \text{(definition of } \notin \text{)} \\ &= \{x \mid \neg(x \in A \wedge x \in B)\} && \text{(definition of intersection)} \\ &= \{x \mid \neg(x \in A) \vee \neg(x \in B)\} && \text{(De Morgan's law for propositions)} \\ &= \{x \mid x \notin A \vee x \notin B\} && \text{(definition of } \notin \text{)} \\ &= \{x \mid x \in \bar{A} \vee x \in \bar{B}\} && \text{(definition of complement)} \\ &= \{x \mid x \in \bar{A} \cup \bar{B}\} && \text{(definition of union)} \\ &= \bar{A} \cup \bar{B} && \blacksquare\end{aligned}$$

REMARK

Set identity 증명이 **propositional equivalence** 로 귀결됩니다.
Week 1에서 배운 논리 법칙이 그대로 집합의 세계에서 활용되는 것!

Proof: De Morgan's Law – Element-wise Method

Claim: $\overline{A \cup B} = \bar{A} \cap \bar{B}$

($\subseteq$) Let $x \in \overline{A \cup B}$ be arbitrary.

$x \notin A \cup B$
 $\Rightarrow x \notin A$ and $x \notin B$ (definition of union)
 $\Rightarrow x \in \bar{A}$ and $x \in \bar{B}$
 $\Rightarrow x \in \bar{A} \cap \bar{B}$

($\supseteq$) Let $x \in \bar{A} \cap \bar{B}$ be arbitrary.

$x \in \bar{A}$ and $x \in \bar{B}$
 $\Rightarrow x \notin A$ and $x \notin B$
 $\Rightarrow x \notin A \cup B$ (definition of union)
 $\Rightarrow x \in \overline{A \cup B}$

Therefore $\overline{A \cup B} = \bar{A} \cap \bar{B}$ ■

REMARK

Logical equivalence method: set-builder로 풀어서 한 방향 chain 으로 충분

Element-wise method: $\subseteq$ 와 $\supseteq$ 양방향 포함 관계를 각각 증명

Inclusion-Exclusion Principle

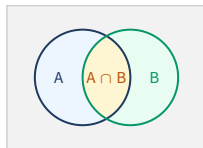
Principle

$$|A \cup B| = |A| + |B| - |A \cap B|$$

$A \cap B$ 를 두 번 더했으므로 한 번 빼야 합니다.

Extension to 3 Sets

$$\begin{aligned} |A \cup B \cup C| &= |A| + |B| + |C| \\ &- |A \cap B| - |A \cap C| - |B \cap C| \\ &+ |A \cap B \cap C| \end{aligned}$$



Example

1-100 중 3의 배수 or 5의 배수?

$$|A| = \lfloor 100/3 \rfloor = 33, |B| = \lfloor 100/5 \rfloor = 20$$

$$|A \cap B| = \lfloor 100/15 \rfloor = 6$$

$$|A \cup B| = 33 + 20 - 6 = \mathbf{47}$$

In C++: popcount

```
popcount(a | b)
= popcount(a) + popcount(b)
- popcount(a & b)
```

Bitmask Inclusion-Exclusion!

Computer Representation: Bit Strings

Idea

$U = \{a_1, a_2, \dots, a_n\}$ 을 순서대로 나열한 뒤,
부분집합 A 를 길이 n 의 bit string 으로 표현: i -th bit = 1 iff $a_i \in A$

Example: $U = \{1, 2, \dots, 10\}$

Set	Bit String
Odd: $\{1, 3, 5, 7, 9\}$	10 1010 1010
Even: $\{2, 4, 6, 8, 10\}$	01 0101 0101
≤ 5 : $\{1, 2, 3, 4, 5\}$	11 1110 0000

Why Bits?

Element listing needs $O(n)$ search for union/intersection, but bit representation enables **hardware-level** single-op processing

Bitwise = Set Operation

$A = \{1, 2, 3, 4, 5\}$: 11 1110 0000

$B = \{1, 3, 5, 7, 9\}$: 10 1010 1010

Union ($A \cup B$):

11 1110 0000 $\vee$ 10 1010 1010
 $= 11 1110 1010 \rightarrow \{1, 2, 3, 4, 5, 7, 9\}$

Intersection ($A \cap B$):

11 1110 0000 $\wedge$ 10 1010 1010
 $= 10 1010 0000 \rightarrow \{1, 3, 5\}$

Complement ($\bar{A}$):

$\neg 11 1110 0000 = 00 0001 1111$

OR $\leftrightarrow \cup$ **AND** $\leftrightarrow \cap$
NOT $\leftrightarrow \neg$ **XOR** $\leftrightarrow \oplus$ (symmetric diff.)

Sets in C++

Bitmask (single integer)

```
// U={1,...,5}, A={1,3,5}, B={1,2,4}
int A = 0b10101; // bit i = elem i
int B = 0b01011;
int U = 0b11111;

int uni  = A | B; // union
int intr = A & B; // intersection
int diff = A & ~B; // difference
int comp = U & ~A; // complement
int sym  = A ^ B; // sym. diff.

// |A| = popcount
int sz = __builtin_popcount(A);

// x in A? (x=3)
bool in = (A >> 3) & 1;
```

std::set

```
#include <set>
set<int> A = {1, 3, 5};
set<int> B = {1, 2, 4};

// insert: A U {7}
A.insert(7);
// erase: A - {3}
A.erase(3);
// x in A?
bool ok = A.count(5) > 0;
// |A|
int sz = A.size();
```

Comparison

Bitmask: $O(1)$, $n \leq 64$
bitset<N>: $O(N/64)$, large N
std::set: $O(\log n)$, no size limit

Multiset

Definition

Multiset은 원소의 중복을 허용하는 집합입니다.

각 원소가 몇 번 등장하는지(**multiplicity**)가 중요합니다.

Notation: $\{a, a, a, b, b\}$ or $\{3 \cdot a, 2 \cdot b\}$ Cardinality: $3 + 2 = 5$

Multiset Operations

$P = \{3 \cdot a, 2 \cdot b\}$, $Q = \{1 \cdot a, 4 \cdot b\}$

Operation	Result
$P \cup Q$ (max)	$\{3 \cdot a, 4 \cdot b\}$
$P \cap Q$ (min)	$\{1 \cdot a, 2 \cdot b\}$
$P - Q$	$\{2 \cdot a, 0 \cdot b\}$
$P + Q$ (sum)	$\{4 \cdot a, 6 \cdot b\}$

In C++: `std::multiset`

```
, 3, 5
// multiplicity of 3 int cnt = ms.count(3);
// erase one copy ms.erase(ms.find(3)); //
// erase all copies of 3 ms.erase(3);
```

REMARK

Sorted + duplicates allowed?
→ `multiset` is the go-to container.

Countable vs Uncountable Sets

Countable Set

원소를 $a_1, a_2, a_3, \dots$ 처럼 자연수와 bijection 시킬 수 있는 집합

e.g. $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ – all countable

Uncountable Set

자연수와 bijection이 불가능한 무한 집합

e.g. $\mathbb{R}$ – proved by Cantor's diagonal argument

CS Motivation

Programs의 집합은 countable (source code = finite string)

Functions $\mathbb{N} \rightarrow \{0, 1\}$ 의 집합은 uncountable

⇒ 프로그램으로 계산할 수 없는 함수가 반드시 존재

⇒ **Halting Problem**의 undecidability 등

이 내용은 Theory of Computation에서 깊이 다루게 됩니다.

Functions – Expressed via Sets

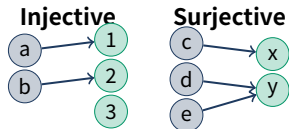
Definition

Function $f : A \rightarrow B$ 는 A의 각 원소에 B의 원소를 **정확히 하나씩** 대응시키는 규칙
수학적으로: $f \subseteq A \times B$, where for each $a \in A$, $\exists! b$ s.t. $(a, b) \in f$

A: **domain** B: **codomain**
 $f(A) = \{f(a) \mid a \in A\}$: **range**

Types of Functions

- **Injective (One-to-One):**
 $f(a) = f(b) \Rightarrow a = b$
distinct inputs $\rightarrow$ distinct outputs
- **Surjective (Onto):**
 $\forall b \in B, \exists a, f(a) = b$
every element in B is mapped to
- **Bijective:** Injective + Surjective
inverse f^{-1} exists



REMARK

Injective: collision-free hash function
Bijective: encryption (f^{-1} = decryption)
In C++: `std::map<A, B>` = mapping

Matrix – Definition

Definition

Matrix는 수를 직사각형으로 배열한 구조입니다.

$m \times n$ matrix: m rows, n columns, entry a_{ij} : row i , column j

Two matrices are equal iff same size and all corresponding entries are equal.

Basic Operations

Addition: same size, entry-wise

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 5 & 5 \end{bmatrix}$$

Multiplication: $A (m \times k), B (k \times n) \rightarrow AB (m \times n)$

$$(AB)_{ij} = \sum_{l=1}^k a_{il} \cdot b_{lj}$$

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 8 & 7 \end{bmatrix}$$

Special Matrices

- **Square:** $m = n$
- **Identity** I_n : diagonal 1, rest 0
 $AI_n = I_m A = A$
- **Transpose** A^T : row $\leftrightarrow$ column
 $(A^T)_{ij} = a_{ji}$
- **Symmetric:** $A = A^T$

REMARK

Matrix multiplication is
NOT commutative: $AB \neq BA$ in general

Matrix Multiplication (Cont.)

$AB \neq BA$ – Example

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$$

$$AB = \begin{bmatrix} 1 \cdot 0 + 2 \cdot 2 & 1 \cdot 1 + 2 \cdot 1 \\ 3 \cdot 0 + 4 \cdot 2 & 3 \cdot 1 + 4 \cdot 1 \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ 8 & 7 \end{bmatrix}$$

$$BA = \begin{bmatrix} 0 \cdot 1 + 1 \cdot 3 & 0 \cdot 2 + 1 \cdot 4 \\ 2 \cdot 1 + 1 \cdot 3 & 2 \cdot 2 + 1 \cdot 4 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 5 & 8 \end{bmatrix}$$

$$AB \neq BA \quad \checkmark$$

Why Matrices Matter in CS

Graph Theory

Adjacency matrix – vertex connectivity

Database

Relation matrix – table correspondence

Computer Graphics

Transformation matrix – rotation, scaling

PS / Competitive

Boolean matrix product – path existence

Machine Learning

Weight matrix – neural network computation

Boolean Matrix (Zero-One Matrix)

Definition

원소가 오직 **0** 또는 **1**인 행렬. Set, Relation, Graph를 행렬로 표현할 때 핵심 도구입니다.

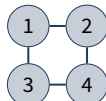
Boolean Operations

- **Join** ($A \vee B$): $c_{ij} = a_{ij} \vee b_{ij}$ (OR)
- **Meet** ($A \wedge B$): $c_{ij} = a_{ij} \wedge b_{ij}$ (AND)
- **Boolean Product** ($A \odot B$):

$$c_{ij} = \bigvee_{k=1}^p (a_{ik} \wedge b_{kj})$$

normal product: $+$ $\rightarrow$ OR, $\times$ $\rightarrow$ AND

Adjacency Matrix



$$M_G = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Boolean Product Example

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$A \odot B = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

REMARK

$M_G^{[2]} = M_G \odot M_G$: (i, j) entry = 1
 $\Leftrightarrow$ path of length 2 from i to j exists
 $M_G^{[r]}$: path of length $\leq r$

Boolean Matrix in C++

bitset adjacency matrix

```
#include <bitset>
const int N = 4;

// adj[i]: neighbor set of vertex i
bitset<N> adj[N];

void addEdge(int u, int v) {
    adj[u].set(v);
    adj[v].set(u); // undirected
}

// Boolean product:  $M^2$ 
//  $M2[i][j]=1$ : path  $i \rightarrow k \rightarrow j$  exists
bitset<N> M2[N];
for (int i = 0; i < N; i++)
    for (int k = 0; k < N; k++)
        if (adj[i].test(k))
            M2[i] |= adj[k];
```

Relation as Matrix

```
// R in  $A \times B$ 
//  $A=\{0,1,2\}$ ,  $B=\{0,1,2,3\}$ 
//  $R=\{(0,1), (1,2), (2,0), (2,3)\}$ 

bitset<4> R[3];
R[0].set(1); // (0,1)
R[1].set(2); // (1,2)
R[2].set(0); // (2,0)
R[2].set(3); // (2,3)

// Matrix:
// [0 1 0 0]
// [0 0 1 0]
// [1 0 0 1]

// Transitive closure:  $R^+$ 
// -> Floyd-Warshall with bitset
```

Performance

bitset Boolean matrix product:
 $O(n^3) \rightarrow O(n^3/64)$ (**64x** speedup)

Application: Transitive Closure

Idea

Relation R 의 **transitive closure** R^+ : 간접 도달 가능한 모든 쌍을 추가
 $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R^+$ (formal definition in Week 3)

Floyd-Warshall + bitset

```
const int N = 500;
bitset<N> reach[N]; // reach[i][j]
                    // if i can
                    // reach j
// Initialize with direct edges
for (auto [u,v] : edges)
    reach[u].set(v);

// Transitive closure
for (int k = 0; k < N; k++)
    for (int i = 0; i < N; i++)
        if (reach[i].test(k))
            reach[i] |= reach[k];

// Now reach[i][j]=1 iff path i->j
```

Intuition

k를 경유지로 삼아:
“i에서 k로 갈 수 있고,
k에서 j로 갈 수 있으면,
i에서 j로 갈 수 있다”

This is Boolean matrix product:
 $\text{reach}[i] \mid= \text{reach}[k]$
(i의 도달 집합에 k의 도달 집합을 합침)

Complexity

$O(N^3/64)$ with bitset
 $N = 500 \rightarrow \text{about } 2 \times 10^6 \text{ ops}$

Week 3 Preview: Relations & Functions

This Week's Tools → Next Week

$A \times B$ (Cartesian Product) → Relation $R \subseteq A \times B$

Boolean Matrix → Matrix representation M_R

Function $f : A \rightarrow B$ → Special case of Relation

Week 3 Topics

Concept	Property
Reflexive	$\forall a, (a, a) \in R$
Symmetric	$(a, b) \in R \Rightarrow (b, a) \in R$
Transitive	$(a, b), (b, c) \in R \Rightarrow (a, c) \in R$
Equivalence	Refl. + Sym. + Trans.
Partial Order	Refl. + Antisym. + Trans.

Example

$A = \{1, 2, 3\}$, relation: “ $a \leq b$ ”

$R = \{(1, 1), (1, 2), (1, 3),$
 $(2, 2), (2, 3), (3, 3)\}$

$$M_R = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

REMARK

Boolean product of M_R computes
Transitive Closure
→ direct application of today's topic!

Basic Track – BOJ Step 49: Set & Map (9 problems)

#	Problem	Key Concept
10815	Numeric Cards	set – membership test ($x \in A?$)
14425	String Set	set<string> – string membership
7785	People in Office	set – insert & erase
1620	Pokémon Master	map – bidirectional mapping
10816	Numeric Cards 2	map / multiset – counting
1764	Never Seen Never Heard	set – intersection ($A \cap B$)
1269	Symmetric Difference	set – $A \oplus B = (A - B) \cup (B - A)$
11478	Distinct Substrings	set – duplicate removal
11403	경로 찾기	Matrix – boolean Matrix

REMARK

오늘 배운 집합 이론이 그대로 STL 설계에 반영되어 있습니다.
set, map, multiset 등 STL container를 자유롭게 활용해보세요.

Practice Track (Cont.)

Advanced Track – BOJ 1717: Disjoint Set (Union-Find)

Link: acmicpc.net/problem/1717

Goal: 집합의 합치기(union)와 같은 집합 판별(find) 구현

Hint: “a와 b가 같은 집합에 속하는가?” → Union-Find (Disjoint Set)
Path compression + union by rank 으로 amortized $O(\alpha(n))$

Connection: 집합의 partition 과 equivalence relation (Week 3 preview)

Challenge Track – BOJ 22940: Linear System of Equations

Link: acmicpc.net/problem/22940

Goal: 행렬 연산을 이용한 연립 방정식 풀기

Hint: n차 연립 방정식을 $AX = B$ 형태의 행렬 곱으로 표현한 뒤,
Gaussian Elimination 으로 해를 구합니다.

Connection: 오늘 배운 행렬 곱의 직접적 활용!

Summary & Week 3 Preview

Week 1

Propositions & Logic

$$p \wedge q$$

$$p \vee q$$

$$p \rightarrow q$$

Logic, SAT

Week 2 – Today

Sets & Matrices

$$A \cap B$$

$$A \cup B$$

$$M_R$$

Sets, Matrix

Week 3

Relations & Functions

$$R \subseteq A \times B$$

$$f : A \rightarrow B$$

$$R^+$$

Relations, Functions

